

```
>g++ campus_navigation.cpp -o campus_navigation.exe
>campus_navigation
***** Student Campus Navigation System *****
```

Type a number to choose an action:

1. Campus Visualization
2. List campus nodes
3. Dijkstra's Algorithm (find the shortest path between two locations)
4. A* Search Algorithm (Uses heuristics to reach the destination faster)
0. Exit

Enter Choice: 1

===== CAMPUS GRAPH VISUALIZATION =====

```
Wilson Library ----120m---- Founders Hall ----140m---- Main Gate
      |                  |
      80m                150m
      |                  |
Hoover Building ----100m---- Abraham Campus Center ----110m---- Parking Garage
      |                  |                  |
      170m               90m                160m
      |                  |                  |
Sports Pavilion      Citrus Hall ----130m---- Vista La Verne
```

Press any key to return to Main Menu!

```
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Enter Choice: 2

===== CAMPUS NODES =====

Node ID Building

ACC Abraham Campus Center
CH Citrus Hall
FH Founders Hall
G1 Main Gate
HB Hoover Building
LIB Wilson Library
PG Parking Garage
SAP Sports Science & Athletics Pavilion
VLV Vista La Verne

Press any key to return to Main Menu!

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0. Exit

Enter Choice: 3

===== DIJKSTRA ALGORITHM =====

Enter Start Node ID : G1

Enter Destination Node ID : VLV

===== DIJKSTRA SHORTEST PATH =====

Start Node : Main Gate

Destination Node : Vista La Verne

Route Graph:

Main Gate --> Parking Garage --> Vista La Verne

Shortest Distance : 340 meters

Estimated Travel Time : 7 minutes

Press any key to return to Main Menu!

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0. Exit

Enter Choice: 3

===== DIJKSTRA ALGORITHM =====

Enter Start Node ID : G1

Enter Destination Node ID : HB

===== DIJKSTRA SHORTEST PATH =====

Start Node : Main Gate

Destination Node : Hoover Building

Route Graph:

Main Gate --> Founders Hall --> Wilson Library --> Hoover Building

Shortest Distance : 340 meters

Estimated Travel Time : 5 minutes

Press any key to return to Main Menu!

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0. Exit

Enter Choice: 3

===== DIJKSTRA ALGORITHM =====

Enter Start Node ID : G1

Enter Destination Node ID : ACC

===== DIJKSTRA SHORTEST PATH =====

Start Node : Main Gate

Destination Node : Abraham Campus Center

Route Graph:

Main Gate --> Founders Hall --> Abraham Campus Center

Shortest Distance : 290 meters

Estimated Travel Time : 5 minutes

Press any key to return to Main Menu!

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0. Exit

Enter Choice: 3

===== DIJKSTRA ALGORITHM =====

Enter Start Node ID : VLV

Enter Destination Node ID : LIB

===== DIJKSTRA SHORTEST PATH =====

Start Node : Vista La Verne

Destination Node : Wilson Library

Route Graph:

Vista La Verne --> Citrus Hall --> Abraham Campus Center --> Hoover Building --> Wilson L

Shortest Distance : 400 meters

Estimated Travel Time : 8 minutes

Press any key to return to Main Menu!

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0. Exit

Enter Choice: 4

===== A* SEARCH ALGORITHM =====

Enter Start Node ID : LIB

Enter Destination Node ID : VLV

===== A* SEARCH RESULT =====

Start Node : Wilson Library

Destination Node : Vista La Verne

Route Graph:

Wilson Library --> Hoover Building --> Abraham Campus Center --> Citrus Hall --> Vista La

Estimated Distance : 400 meters

Estimated Travel Time : 8 minutes

Press any key to return to Main Menu!

***** Student Campus Navigation System *****

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2. List campus nodes
3. Dijkstra's Algorithm (find the shortest path between two locations)
4. A* Search Algorithm (Uses heuristics to reach the destination faster)
0. Exit

Enter Choice: 4

===== A* SEARCH ALGORITHM =====

Enter Start Node ID : G1
Enter Destination Node ID : SA

ERROR: Invalid Node ID!
Please enter a valid Node ID.

Enter Destination Node ID : SAP

===== A* SEARCH RESULT =====

Start Node : Main Gate
Destination Node : Sports Science & Athletics Pavilion

Route Graph:

Main Gate --> Sports Science & Athletics Pavilion

Estimated Distance : 200 meters
Estimated Travel Time : 4 minutes

Press any key to return to Main Menu!
***** Student Campus Navigation System *****

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0. Exit

Enter Choice: 0

Thank you for using the Student Campus Navigation System! Goodbye!